Sustainable Investment Circle – Candidate Infrastructure Architecture

Member community platform – application, identity, data, communication & commercial architecture · **candidate only**: layers marked with a ticket are decided on the Wayfinder map

1. Access & Experience

who uses it

Members
ESG & sustainable-investment professionals · free & Premium

Community team
Admins / moderators · approvals · support inbox

Partners & advertisers
Partner profiles · placements

Phone camera
Scans a member's QR → opens their profile in the browser

Web browser
Any device · responsive web (no native app in v1)
`#25 responsive`

HTTPS / TLS · secure connection

2. Source Control & Deployment Layer

GitHub – Business-Analyst-Services/sustainable-investment-circle
Source code · Wayfinder decision map (Issues) · prototype on GitHub Pages · CI/CD via Actions

Environments

main	preview	production
development	staging / UAT	live members

Code push / CI-CD / Deployments

SIC Platform

Custom domain / DNS
Member-facing domain · TLS certificate
`#29 accounts-setup`

Vercel – SIC Web App

Frontend (Next.js) · edge hosting & delivery · Sydney region (syd1)

Home & feed · Members directory · Channels · Messages · Events · Research · Partners · Profile · Premium · Admin console

`#3 build-vs-platform` `#8 tech-stack`

Login / Session / Identity
protected routes · role-based access

3. Identity & Security Layer

Clerk

Authentication · sessions · user management

email + password / magic link · MFA · roles: member · premium · admin · partner · corporate-email verification?

`#9 identity-auth` `#5 eligibility`

Requests ↓ · Responses ↑

4. Application / API Layer

Application Logic / Server Actions / API

Business rules · connection & visibility rules · moderation hooks · Premium entitlement checks · search & filters

connections · messaging · feed rules · approvals · who-viewed-me

`#10 domain-model` `#11–13` `#16`

Generated artefacts

Profile **PDF** (serverless render) · connection **QR** codes · **ICS** calendar files

`#18 profile-pdf-qr` `#20 events`

Read / write data ↓ · Send emails

5. Data Layer

Neon

Postgres · Sydney (ap-southeast-2) · branching per preview
`#6 privacy / residency`

Members & orgs profiles · roles

Connections requests · graph

Channels & posts feed · reactions

Messages DMs · groups

Events · providers · partners

Subscriptions & profile views

Vercel Blob / S3

Photos · post attachments (5 × 25 MB) · uploaded documents

Search: Postgres full-text first; search service only if scale demands `#14`

6. Communication Layer

Resend

Transactional email · verified sending domain

welcome / verify · connection requests · message alerts · event reminders · Premium search alerts · invites (CSV) · digests

`#23 notifications` `#22 invite-import`

Commercial Layer

Airwallex

External payment link – no card data in the app. US$99 founding / US$149 Premium, annual.

request → pay → activate · manual vs webhook · renewal · lapse · GST

`#15 premium-entitlement` `#4 commercial`

7. Observability / Operations

run & protect

Analytics / monitoring
Vercel Analytics · success metrics at 30/90/180 days · Premium conversion
`#26 metrics`

Logs / audit trail
Application logs · moderation & approval history · profile-view records (consented)
`#16 who-viewed-me` `#24 admin`

Env vars / secrets · backups
Vercel env per environment · Neon PITR backups · breach-notification runbook
`#6 privacy`

Policies & legal
Privacy Policy · Terms · Community Guidelines · Partner & Premium T&Cs
`#28 legal-docs`

8. Later / Optional integrations

not in the v1 destination unless a ticket pulls them in

Video hosting
YouTube / Vimeo embed for the featured community video

Jira mirror
Implementation tickets after /to-tickets

CRM / outreach
Founding-member campaign · partner pipeline `#27`

Payments in-app
Out of scope for v1 – Airwallex link only

AI recommendations
Out of scope for v1 – "people to connect with" is rules-based

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request / data flow · telemetry · config · reference · `#n` Wayfinder ticket that decides this element

Stack mirrors the BAS platform (GitHub → Vercel · Clerk · Neon · Resend) as the default candidate; #3 build-vs-platform may replace it with a community platform.